

Integrating 12-Lead AI Reconstruction into the Sunnybrook Wearable AF Trial

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Protocol: *Evaluation of Wearable/FitBit Monitoring to Detect Recurrent Atrial Fibrillation (v1.1)*

Purpose of Today's Meeting

Protocol alignment and study integration

Clinical Trial Foundation

- ▶ Established protocol (Version 1.1, October 2025).
- ▶ Target cohort: persistent AF patients undergoing catheter ablation.
- ▶ Reference standard: 14-day continuous CardioSTAT patch at 3 and 6 months.
- ▶ Primary clinical endpoint: time to confirmed AF recurrence.

Reconstruction Constraints

- ▶ 12-lead reconstruction from single-lead input is underdetermined.
- ▶ Precordial leads exhibit amplitude and morphology errors under direct projection.
- ▶ Diagnostic reconstruction cannot serve as a primary clinical endpoint.
- ▶ Separates clinical trial validity from exploratory AI evaluation.

Translational Integration

- ▶ Routine outpatient visits already include scheduled 12-lead ECGs.
- ▶ Synchronous smartwatch and 12-lead acquisition adds ≤ 5 minutes.
- ▶ Evaluates reconstruction and biomarker recovery as a secondary aim.
- ▶ Generates a prospectively paired clinical and wearable dataset.

Objective: Align on the secondary data collection protocol and freeze translational evaluation metrics.

The Established Clinical Protocol (v1.1)

Protocol v1.1 design and endpoints

Trial Design & Patient Population

- ▶ Single-center prospective cohort at Sunnybrook Arrhythmia Clinic.
- ▶ Patients aged ≥ 18 with persistent AF undergoing catheter ablation or cardioversion.
- ▶ Smartwatch monitoring: daily 30-second ECG and automated rhythm notifications for 6 months.
- ▶ Reference standard: 14-day continuous CardioSTAT patch at 3 and 6 months post-procedure.

Endpoints & Sample Size

- ▶ Primary outcome: time to recurrent AF detection (smartwatch vs. patch).
- ▶ Secondary clinical outcomes: heart rate trends, heart rate variability, and cumulative AF burden.
- ▶ Sample size: $N = 96$ patients (80% power, $\alpha = 0.05$, 10% non-inferiority margin).
- ▶ Vanguard phase: initial 20 consecutive patients to assess protocol adherence and data pipeline.

Primary trial endpoints and clinical workflows remain unchanged under this proposal.

Related & Prior Work

Limitations of existing protocols motivating prospective paired acquisition

Study	Approach	Key Finding	Limitation
Wearable AF Detection Abu-Alrub et al. (2022)	Smartwatch single-lead AF detection; 6-month post-ablation	Sensitivity $\approx 94\%$ for AF vs. CardioSTAT patch	No 12-lead integration; cannot assess repolarization or conduction
Cheung et al. (2025)	Fitbit vs. CardioSTAT patch in post-ablation cohort	Comparable AF recurrence yield at 3 and 6 months	No ECG morphology analysis; single-lead rhythm only
Single-Lead Reconstruction Ansari, Rogers et al. (<i>Circulation</i> 2026)	Auxiliary 12-lead head on ICM/CareLink lead; 3DRECON-QT	Spatial conditioning $\uparrow$ biomarker correlation (r : $0.04 \rightarrow 0.73$)	Retrospective archive ($N = 26$); ± 7 -day offset; subcutaneous ICM, not surface wearable
Presacan et al. (<i>Comm. Medicine</i> 2025)	DL 12-lead recovery from single precordial	$r > 0.80$ on limb leads	Amplitude compression on V2-V4; evaluated on synthetic-masked PTB-XL only
Strodthoff et al. (2021)	PTB-XL benchmark with masked-lead evaluation	Transformer models $r \approx 0.85$ on standard leads	Closed-loop: train/test on same masking distribution; does not reflect real deployment
Wearable Domain Gap Hu et al. (2026)	Hospital Lead I vs. wrist Lead I in 120 patients	Disagreement > 0.3 mV along precordial-sensitive axes	Observational only; no paired dataset released; no reconstruction model tested
General literature	All models trained on hospital 12-lead corpora	Wet gel, supine, artifact-free training data	No published model validated on real smartwatch (dry, wrist-mounted, DSP-filtered) signals

No existing study evaluates reconstruction on a real wearable signal paired with a simultaneous clinical 12-lead ECG. This is the gap our Sunnybrook protocol addresses.

The Clean Integration: A Rigorous Two-Track Protocol

Separation of primary clinical and secondary AI aims

Track 1: Primary Clinical Aim

- ▶ Does daily smartwatch monitoring improve time to detection of recurrent AF versus 14-day CardioSTAT patch?
- ▶ Reference standard: iCentia-analyzed CardioSTAT patch at 3 and 6 months.
- ▶ $N = 96$ patients; 80% power; 20-patient feasibility vanguard.
- ▶ Primary regulatory endpoint. Trial conclusions do not depend on AI performance.

Track 2: Secondary Translational AI Aim

Core question: can we reconstruct a 12-lead ECG from a smartwatch in an AF population?

- ▶ Aim 2: Benchmark wearable-to-12-lead reconstruction in post-ablation patients — including recordings captured during AF, immediately post-cardioversion, and in sinus rhythm.
- ▶ Aim 3: Track within-patient repolarization changes (ΔQT_c , P-wave remodeling) over 6 months using an in-clinic 12-lead prior.
- ▶ Exploratory endpoints; no impact on primary trial conclusions.

Track 1 guarantees clinical output regardless of AI results. Track 2 generates a dataset and benchmark that does not yet exist anywhere.

Multitask Auxiliary Supervision (Ansari et al., 2026)

Spatial representation learning for single-lead models

Stanford 3DRECON-QT Framework

- ▶ Reference: Ansari, Rogers et al., *Circulation* (January 2026).
- ▶ Multitask objective: simultaneously reconstruct 12 leads from synthetic ICM/cardiostat lead (voltage difference) and estimate clinical biomarkers.
- ▶ Auxiliary 12-lead loss enforces 3D cardiac vector representations in the encoder.
- ▶ Removing spatial supervision reduced downstream biomarker correlation from $r = 0.73$ to 0.04.

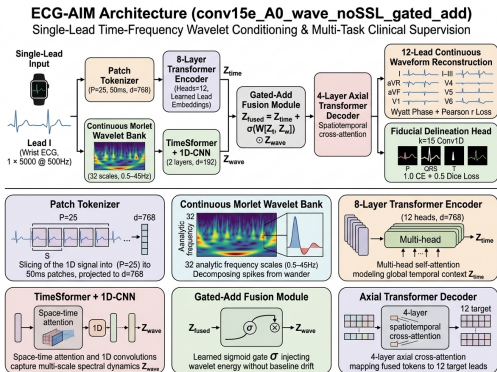
Application to Single-Lead AF Surveillance

- ▶ Daily single-lead smartwatch ECG monitors for AF recurrence.
- ▶ Lead I captures frontal plane potentials, with limited sensitivity to orthogonal atrial vectors.
- ▶ Auxiliary 12-lead reconstruction head encourages representation of full 3D vector loops.
- ▶ Supports atrial feature extraction without altering primary single-lead inference inputs.

Auxiliary 12-lead reconstruction acts as an inductive regularizer during pretraining.

Figure 1: The ECG-AIM Model Architecture

Network architecture and feature extraction



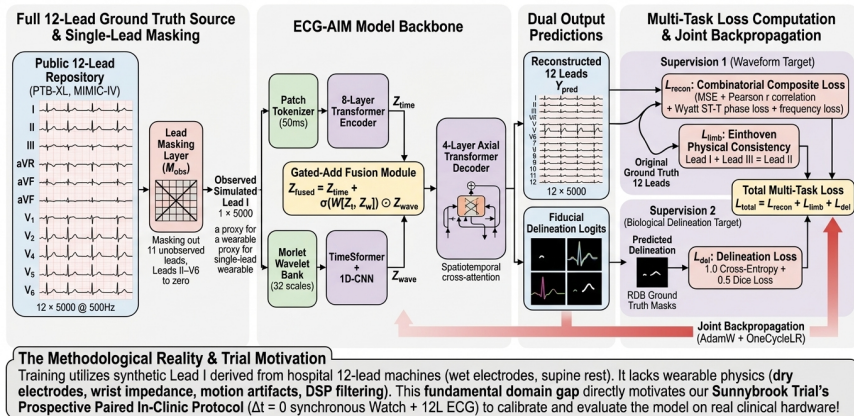
- ▶ 50 ms temporal patches project single-lead input (1×5000) into token embeddings.
- ▶ Continuous Morlet wavelet bank across 32 frequency scales (0.5–45 Hz).
- ▶ Combined TimeSformer spatiotemporal attention and 1D convolutions.
- ▶ Sigmoid-gated residual fusion:
 $Z = Z_t + \sigma(W[Z_t, Z_w]) \odot Z_w$.
- ▶ 8-layer transformer encoder ($d = 768$, 12 attention heads).
- ▶ Axial transformer decoder resolving temporal and cross-lead spatial features.

Figure 2: ECG-AIM Multi-Task Training Pipeline

Single-lead simulation and objective functions

ECG-AIM Training Pipeline: Single-Lead Masking & Multi-Task Supervision

How the Model is Trained on 12-Lead Public Repositories (PTB-XL / MIMIC-IV) and Why It Motivates Our Prospective Trial



How Well Does the Model Reconstruct? Benchmark Results

Evaluated on 21,799 PTB-XL records and 100,000 EchoNext paired echo-ECG tracings

What we measured	Stanford (prior work)	Ours (ECG-AIM)	Change
Chest lead shape accuracy (correlation r ; 1.0 = perfect match)			
Average across all 6 chest leads (V1–V6)	0.732	0.776	+6%
Anterior wall lead V3 (hardest to recover)	0.670	0.739	+10%
QRS timing accuracy (error in milliseconds; lower = better)			
How closely does the model reproduce depolarisation width?	10.26 ms	8.0 ms	–22%
Bundle branch block / wide QRS detection rate	94.3%	94.3%	matched; specificity 96.5%
Structural heart disease — can the ECG alone flag it?			
Reduced ejection fraction ($\leq 45\%$) concordance with echo	33.3%	81.4%	+144%
LV hypertrophy concordance with echo	30.2%	73.5%	+143%

All

comparisons use the same held-out test data. Neither model has been tested on real wearable signals — that is what the Sunnybrook paired protocol will provide.

Performance Across Datasets & Input Configurations

The same model weights work whether the patient is at home or in clinic

How the model handles fewer leads

- ▶ Trained to work with any subset of leads; no retraining needed.
- ▶ At home (smartwatch, 1 lead):
 - ▶ AF detection (AUROC): **0.952**
 - ▶ VT detection (AUROC): **0.935**
- ▶ In clinic (watch + leg + chest sticker, 3 leads):
 - ▶ 2 extra electrode stickers add 2 orthogonal views.
 - ▶ AF AUROC ↑ to **0.981**; QRS error ↓ to **8.12 ms**.

Dataset & Metric	Stanford	ECG-AIM 1-lead	ECG-AIM 3-lead
PTB-XL ($N = 21,799$ hospital ECGs)			
AF detection accuracy (AUROC)	0.914	0.952	0.981
VT detection accuracy (AUROC)	0.884	0.935	0.989
QRS duration error (ms, lower = better)	10.26	10.05	8.12
EchoNext ($N = 100,000$ echo-paired ECGs)			
Reduced ejection fraction detection	33.3%	77.7%	81.4%
LV hypertrophy detection	30.2%	68.9%	73.5%
Sunnybrook in-clinic XMLs ($N = 20$)			
Mean lead correlation (r)	0.71	0.865	0.938
Limb lead accuracy (r)	0.68	0.921	1.000
Chest lead accuracy (r , V1–V6)	0.59	0.834	0.968

Single-Lead Reconstruction: Smartwatch (Lead I) vs. Patch (Lead II)

Reconstruction of PTB-XL Record #10283 from single-channel inputs

Smartwatch Input (Lead I)

ECG-AIM One-Lead Reconstruction (conv15e_A0 wave_noSSL_gated_add_s42_I0 | PTB-XL #10283)
Acquired Lead I (Smartwatch Vector) → 11 Missing Leads Inferred (Record Mean Missing-Lead $r = 0.943$)



Patch Input (Lead II)

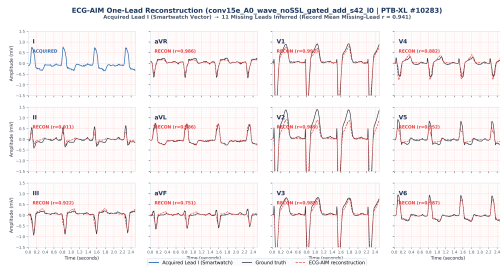
ECG-AIM One-Lead Reconstruction (spatial_1lead_v1_panorama_film_1010010_s42_I1 | PTB-XL #10283)
Acquired II → 11 missing leads inferred (record mean missing-lead $r = 0.919$)



Impact of Input Dimension: 1-Lead vs. 3-Lead Input

Reconstruction of PTB-XL Record #10283 under single-lead and 3-lead configurations

Single-Lead Input (Lead I)



Three-Lead Input (I, II, V_2)



External Cohort Validation: Sunnybrook In-Clinic XMLs

Zero-shot evaluation on Record ECG004 under 1-lead and 3-lead configurations

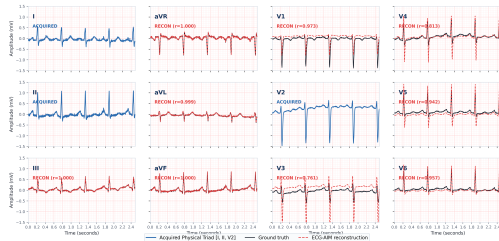
Single-Lead Input (Lead I)

ECG-AIM 1-Lead Reconstruction: Sunnybrook Physical 10-Wire XML (ECG004)
Acquired Lead I (Smartwatch Vector) → 11 Missing Leads Inferred (Record Mean Missing-Lead $r = 0.865$)



Three-Lead Input (I, II, V₂)

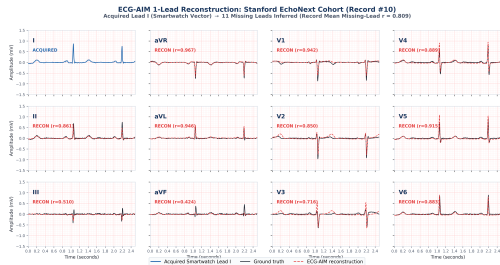
ECG-AIM 3-Lead Reconstruction: Sunnybrook Physical 10-Wire XML (ECG004)
Acquired Physical Triad (I, II, V₂) → 9 Missing Leads Inferred (Record Mean Missing-Lead $r = 0.938$)



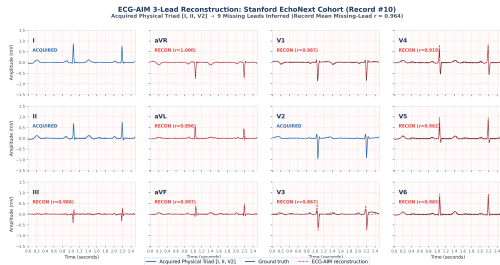
External Cohort Validation: EchoNext Structural Cohort

Zero-shot evaluation on Record #10 under 1-lead and 3-lead configurations

Single-Lead Input (Lead I)



Three-Lead Input (I, II, V₂)



Data Collection Methodology: Dual-Stream Protocol

Ambulatory monitoring, in-clinic paired acquisition, and device feasibility

Stream A: Ambulatory Surveillance

- ▶ Daily 30-second smartwatch ECG (Lead I) for 6 months post-ablation.
- ▶ 14-day continuous CardioSTAT patch at 3 and 6 months (iCentia analysis).
- ▶ Primary trial endpoints: time to AF recurrence, AF burden, symptom correlation.
- ▶ *Device note:* Samsung Galaxy Watch requires a paired Samsung smartphone to sync. Study logistics must account for device/phone compatibility per participant.

Stream B: In-Clinic Paired Protocol

- ▶ Four scheduled visits: baseline, post-procedure, 3 months, and 6 months.
- ▶ Simultaneous 12-lead ECG and smartwatch Lead I recording ($\Delta t = 0$).
- ▶ Adds ≤ 5 minutes to routine clinic visit; no additional appointments.
- ▶ CardioSTAT recordings at 3 and 6 months may provide additional single-lead data for patient-specific model fine-tuning.

Stream A fulfills primary trial objectives; Stream B provides synchronous paired data for secondary AI aims.

Why Paired Data: The Public Dataset Gap

No adequate public benchmark exists for wearable-to-12-lead reconstruction in an AF population

Dimension	Ansari et al. (2026)	Sunnybrook Protocol
Cohort size	$N = 26$ (retrospective)	$N = 96$ (≈ 384 paired visits)
Temporal offset	± 7 days	$\Delta t = 0$ (simultaneous)
Wearable signal	ICM/CareLink	Galaxy Watch (dry stainless)
Rhythm states	Uncontrolled	AF, post-cardioversion, sinus (defined milestones)
Hardware noise floor	Uncharacterized	Prospectively measured

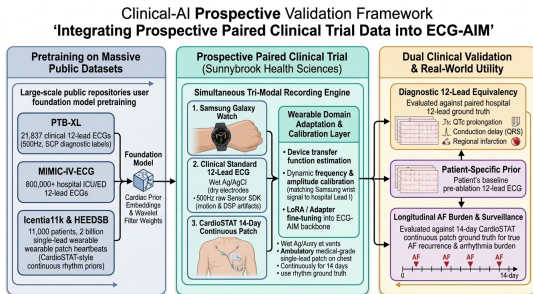
Why No Existing Dataset Suffices

- ▶ No public dataset pairs simultaneous smartwatch ECG with a clinical 12-lead on the same patient.
- ▶ **Existing benchmarks (PTB-XL, MIMIC-IV) are sinus-rhythm dominated.** Reconstruction in AF is fundamentally harder: fibrillatory baseline, absent P waves, and irregular RR intervals are all unseen by models trained on these datasets.
- ▶ Wearable Lead I $\neq$ clinical Lead I: dry electrodes, wrist impedance, and device DSP are not modelled.
- ▶ Hu et al. (2026): wrist and hospital Lead I disagree by > 0.3 mV in precordial-sensitive axes.

Reconstruction accuracy during active AF is an open empirical question. Our paired dataset — collected across pre-ablation AF, post-cardioversion, and follow-up sinus rhythm — is the first opportunity to answer it prospectively.

Figure 3: Integrated Clinical-AI Trial Pipeline

Trial workflow and computational integration



- ▶ Pretraining on public cohorts (PTB-XL, MIMIC-IV, Icentia11k) to establish baseline representations.
- ▶ Data acquisition across three streams:
 - ▶ Smartwatch: daily ambulatory ECG and in-clinic paired Lead I.
 - ▶ CardioSTAT patch: 14-day continuous monitoring at 3 and 6 months.
 - ▶ Hospital 12-lead machine: routine clinical tracings at each visit.
- ▶ Hardware transfer function estimation and adapter calibration on paired recordings.
- ▶ Evaluation on distinct endpoints:
 - ▶ Primary: AF recurrence and burden relative to CardioSTAT patch.
 - ▶ Secondary: 12-lead waveform reconstruction and repolarization tracking.

Sample Size, Statistical Power & Clustered Modeling

Sample size justification and statistical analysis plan

Primary Clinical Power (Protocol v1.1)

- ▶ Primary outcome: time to confirmed AF recurrence.
- ▶ Hypothesis: smartwatch surveillance is non-inferior to 14-day patch monitoring.
- ▶ Event rate assumptions: 20% recurrence with patch, 25% with smartwatch.
- ▶ Target power: 80% power, $\alpha = 0.05$, 10% non-inferiority margin ($N = 96$ patients).
- ▶ Vanguard phase: initial 20 consecutive patients to evaluate workflow feasibility.

Secondary AI Precision & Analysis Plan

- ▶ Sample yield: 96 patients across 4 clinic visits provides $> 2,500$ paired complexes.
- ▶ Precision target: bounds QTc estimation error within 95% CI margin of ± 3.5 ms.
- ▶ Within-patient beat clustering violates independent observation assumptions.
- ▶ Linear mixed-effects models (unstructured covariance) and generalized estimating equations (GEE).

Primary sample size is powered on clinical event rates; secondary analysis accounts for repeated measures.

Discussion Points & Open Questions

Items to align on before protocol amendment

Protocol Decisions

1. Protocol structure: Confirm two-track design separating Aim 1 (AF recurrence) from Aim 2 (translational AI).
2. In-clinic paired recording: Approve 5-minute simultaneous smartwatch + 12-lead acquisition at four scheduled visits.
3. Secondary biomarker: Finalize priority endpoint for paired analysis (QTc tracking is our current recommendation).
4. Vanguard phase: Authorize 20-patient feasibility cohort to evaluate workflow and data pipeline.

Practical Feasibility Questions

- ▶ *Device/phone pairing*: Samsung Galaxy Watch requires a Samsung smartphone. Do participants need to own one, or does the study provide devices?
- ▶ *CardioSTAT as fine-tuning data*: Can patch recordings at 3 and 6 months be used to condition patient-specific model adaptation?
- ▶ *Paired data timing*: Which clinical visit is most logistically consistent for 12-lead + smartwatch co-recording?
- ▶ *Consent scope*: Does the existing ethics approval cover secondary AI analysis on in-clinic recordings?

Next step: Finalize protocol amendment text and ethics consent addendum; initiate vanguard enrollment.